Chibuike E. Ugwu

Curriculum Vitae

School of Electrical Engineering and Computer Science,
Washington State University,
Pullman, WA, USA.

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Research Summary

My general research interests are in **Artificial Intelligence (AI)** and **Machine Learning (ML)**, with a main focus on developing robust, safe, and trustworthy ML algorithms with theoretical guarantees for deployment in critical real-world challenges within safety-sensitive domains (e.g., healthcare). My current work focuses on:

- Building trustworthy Large Language Model (LLM) hallucination control policies.
- Advancing **conformal prediction**-based uncertainty quantification with theoretical guarantees to facilitate effective human–ML collaboration (e.g., Clinician-in-the-Loop Uncertainty-aware predictive systems) and beyond.
- Developing uncertainty-aware energy management methods for wearable Internet of Things (IoT) devices for mobile health applications and beyond.

Education

Ph.D., Computer Science (GPA 3.86) — Jul., 2022 – Present
Washington State University — *Pullman, WA, USA*
Advisors: Prof. Jana Doppa
Thesis: Developing Novel Algorithms and Theory for Robust and Trustworthy Machine Learning.

M.Sc., Computer Science — Aug., 2022 – May, 2025
Washington State University — *Pullman, WA, USA*
***Thesis**: Conformalized Uncertainty Regions for Machine Learning-Based Multiple Cognitive Health Measures from Smartwatch Sensor Data.*

B.Sc., Statistics (First Class Honors) — Sept., 2014 – Nov., 2018
University of Nigeria — *Nsukka, Nigeria*
***Thesis**: Sensitivity of Randomization test and F-test in Some Experimental Designs*
*Class rank: **Best Graduating Student***

Awards and Honors

- **Overall Outstanding Research Assistant VCEA** — 2026
  Voiland College of Engineering, Washington State University
- **Outstanding Research Assistant in Computer Science EECS Award** — 2026
  Voiland College of Engineering, Washington State University
- **GPSA Research Assistant of the Year Award** — 2026
  Graduate and Professional Student Association (GPSA), Washington State University
- **AAAI Student Scholarship and Volunteer Program** — 2026
  Association for the Advancement of Artificial Intelligence Conference
- **Mahmoud M. Dillsi Graduate Fellowship** — 2025
  School of Electrical Engineering and Computer Science, Washington State University
- **Nakahara Tsuyoshi and Mary Fellowship** — 2024
  School of Electrical Engineering and Computer Science, Washington State University
- **Outstanding Graduate Teaching Assistant in EECS Award** — 2024
  Voiland College of Engineering, Washington State University

🏆 **EducationUSA Opportunity Fund Program (OFP) Scholar.** 2022
USA Embassy, Abuja, Nigeria

🏆 **Best Graduating Student** 2018
Statistics Department – University of Nigeria

🏆 **Winner: Nigerian Statistical Association (NSA) competition** 2018
Representative of the University of Nigeria.

🏆 **Gold medalist: NSA competition individual category** 2018
Representative of the University of Nigeria.

Professional Appointments

Research Assistant Aug 2022 – Present
EECS Department – Washington State University *Pullman, WA, USA*
– Developing Novel Algorithms and Theory for Robust and Trustworthy Machine Learning.

Teaching Assistant and Guest Lecturer Aug 2022 – Dec 2024
EECS Department – Washington State University *Pullman, WA, USA*
– CptS 223: Advanced Data Structures C/C++ (Fall-2022)
– CptS 315: Introduction to Data Mining (Spring-2023)
– CptS 223: Advanced Data Structures C/C++ (Fall-2023)
– CptS 315: Introduction to Data Mining (Spring-2024)
– CptS 223: Advanced Data Structures C/C++ (Fall-2024)

Publications

1. [**AAAI'26**] Chibuike E. Ugwu, Roschelle Fritz, Diane J. Cook, and Janardhan Doppa. **Clinician-in-the-Loop Smart Home System to Detect Urinary Tract Infection Flare-Ups via Uncertainty-Aware Decision Support**. *40th AAAI Conference on Artificial Intelligence (AAAI)* 2026.
2. [**ACM HEALTH'26**] Chibuike E. Ugwu, Yan Yan, Diane J. Cook, Maureen Schmitter-Edgecombe, and Janardhan Doppa. **Importance-Weighted Calibrated Region Prediction of Multi-target Cognitive Health Measures from Smartwatch Sensor Data**. *ACM Transactions on Computing for Healthcare*, 2026. *Impact factor: 8.0*
3. [**ACM TODAES'26**] Chibuike E. Ugwu*, Dina Hussein*, Ganapati Bhat, and Janardhan Doppa. **Trading Off Performance and Sustainability in Internet of Things: An Uncertainty-Aware Hierarchical Energy Management Approach**. *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2026. (* denotes equal contribution)
4. [**IJCAI'25**] Chibuike E. Ugwu*, Dina Hussein*, Ganapati Bhat, and Janardhan Doppa. **Sustainable Wearables for Health Applications and Beyond via Uncertainty-Aware Energy Management.** *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, 2025 (* denotes equal contribution)
5. [**DAC'25**] Chibuike E. Ugwu*, Dina Hussein*, Ganapati Bhat, and Janardhan Doppa. **Uncertainty-Aware Energy Management for Wearable IoT Devices with Conformal Prediction.** *Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC)*, 2025. (* denotes equal contribution)
6. [**ACM TECS'25**] Pratyush Dhingra, Chibuike E. Ugwu, Janardhan Rao Doppa, and Partha Pratim Pande. **ERGo: Energy-Efficient Hybrid Graph Neural Network Training on Heterogeneous Processing-In-Memory Architecture.** *ACM Transactions on Embedded Computing Systems (TECS)*, 2025.
7. [**QREI'23**] Abimibola V. Oladugba, Chibuike E. Ugwu, and Uchenna C. Onwuamaeze. **Sensitivity and robustness of randomization test and F-test in some experimental designs.** *Quality and Reliability Engineering International*, 39(7), 2967-2974, 2023.

Software Prototypes

Adaptive Prediction Region (APR) [code] Aug., 2024
– Developed a novel framework (APR) for constructing trustworthy, adaptive uncertainty regions for multi-target regression tasks.

Uncertainty Regions via Importance-weighted Calibration (URIC) [code] Nov., 2025
– Designed and implemented software to produce calibrated predictions of multiple clinical cognitive health measures from continuous smartwatch sensor data.

Clinician-in-the-Loop Conformal-Calibrated Intervals (CCI) [code] Jan., 2026
– Built a clinician-facing prototype that surfaces calibrated uncertainty for UTI risk detection and supports human feedback in the decision loop.

Conference Activities

1. AAAI Conference on Artificial Intelligence **(AAAI)** 2026
2. Annual Conference on Neural Information Processing Systems **(NeurIPS)** 2025
3. International Joint Conference on Artificial Intelligence **(IJCAI)** 2025

Professional Services and Outreach Activities

Program Committee Member

1. International Conference on Machine Learning **(ICML)** 2026
2. International Conference on Uncertainty in Artificial Intelligence **(UAI)** 2026
3. International Conference on Learning Representations **(ICLR)** 2026
4. Association for the Advancement of Artificial Intelligence **(AAAI)** 2026
5. AAAI Conference on Artificial Intelligence, AI for Social Impact Track **(AISI)** 2026
6. AAAI Conference on Artificial Intelligence, AI for Innovative Applications **(IAAI)** 2026
7. International Conference of Machine Learning **(ICML)** 2025
8. Association for the Advancement of Artificial Intelligence **(AAAI)** 2025
9. AAAI Conference on Artificial Intelligence, AI for Social Impact Track **(AISI)** 2025
10. AAAI Conference on Artificial Intelligence, AI for Social Impact Track **(AISI)** 2024

Volunteer and Outreach Activities

1. Volunteer for AAAI Conference on Artificial Intelligence (AAAI), 2026
2. Volunteer for International Joint Conference on Artificial Intelligence (IJCAI), 2025
3. Mentor and Judge for Digital AgAthon (AgAID Institute), 2025
4. Instructor for WSU Summer Programming Camp for Middle Schoolers, 2025
5. Judge for Showcase for Undergraduate Research and Creative Activities (SURCA), 2026
6. Judge for ACM Club's CrimsonCode Hackathon, 2026